Rational Functions: Full Practice Exam

Name: _____ Date: _____

13 questions | 58 points | 60 minutes

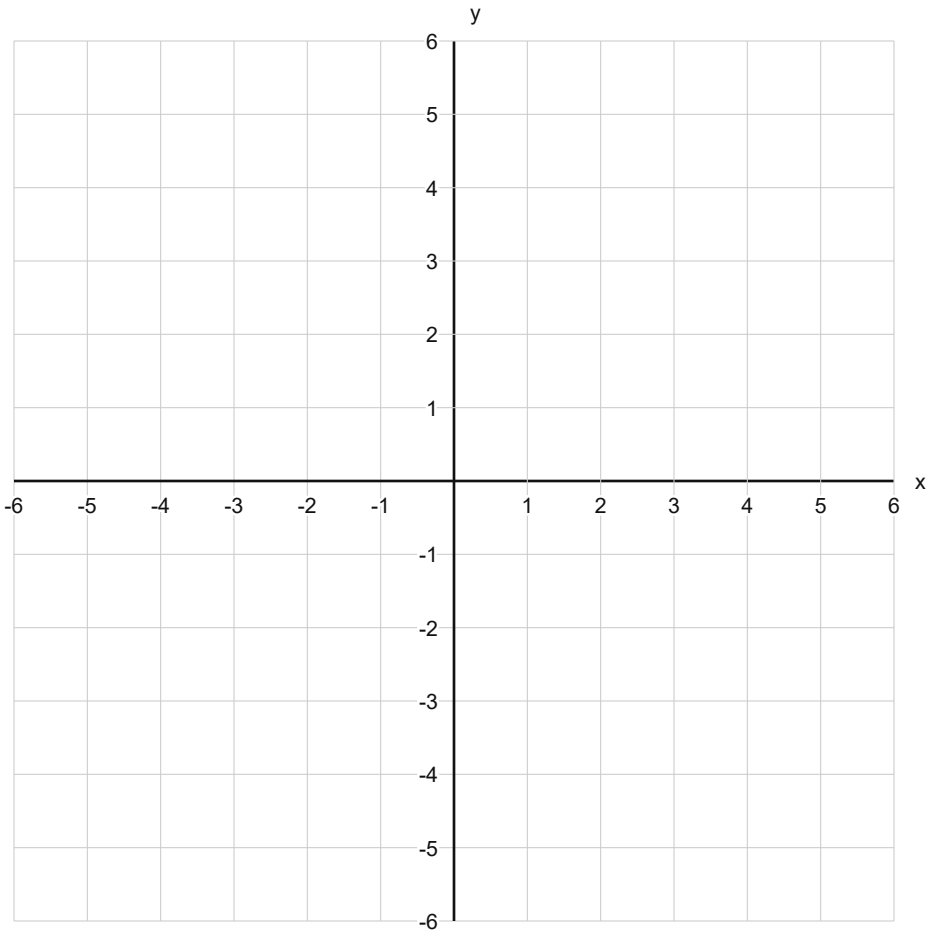
Complete every labeled part. Show the work and explanations requested. Give exact answers unless an estimate is requested. A four-function calculator is permitted.

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1. 8 points

Let $f(x) = 3 - \dfrac{2}{x+1}$.

(a) Graph the function. Label its vertical and horizontal asymptotes and at least two points on the curve.



(b) State the far-left and far-right end behavior.

Far-left behavior	Far-right behavior

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2. 4 points

Simplify $\dfrac{3}{x+2} - \dfrac{2x-1}{x^2-4}$. State all excluded real inputs. Show your algebra.

Work and final answer

Algebra
Simplified expression
Excluded real inputs

3. 4 points

Simplify $\dfrac{x^2-2x-15}{x^2-25} \cdot \dfrac{x+5}{x^2-9}$. State all excluded real inputs. Show your algebra.

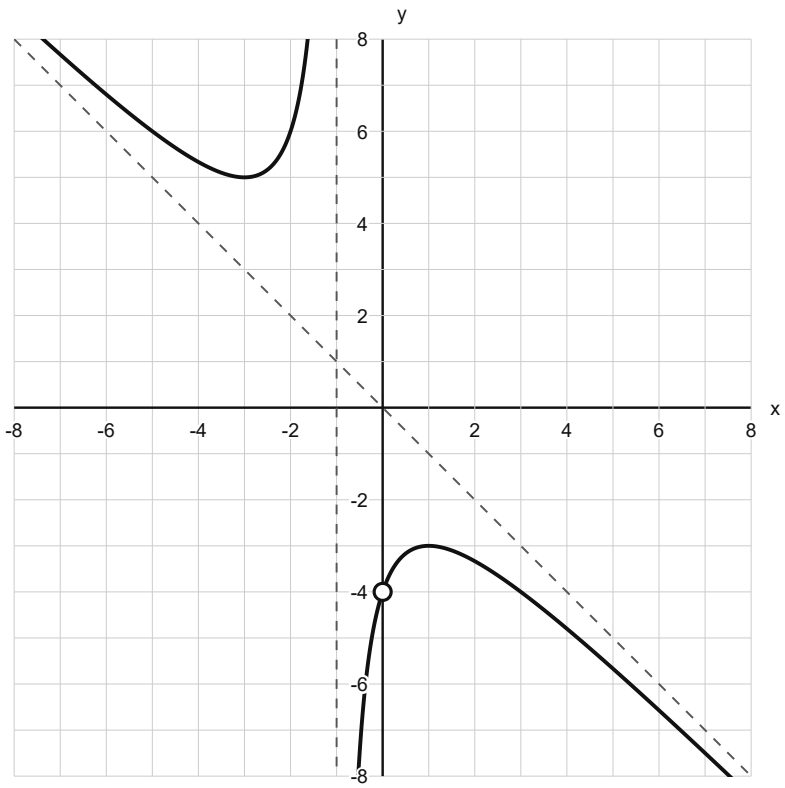
Work and final answer

Algebra
Simplified expression
Excluded real inputs

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4. 8 points

Use the supplied graph to answer each part. The curve continues beyond the window; an open circle denotes a missing point.



(a) Give the intervals where the function is increasing.

(b) Give the coordinates of every relative maximum and relative minimum.

Relative maximum
Relative minimum

(c) Give the intervals where the function is concave up and concave down.

Concave up
Concave down

(d) For each asymptote type, give its equation or state that none exists.

Vertical
Horizontal
Slant

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5. **10 points**

Let $R(x) = \dfrac{2x^2 + 6x + 4}{x^2 - x - 2}$.

(a) Find all intercepts, holes and asymptotes. Give exact coordinates or equations as appropriate and show your algebra.

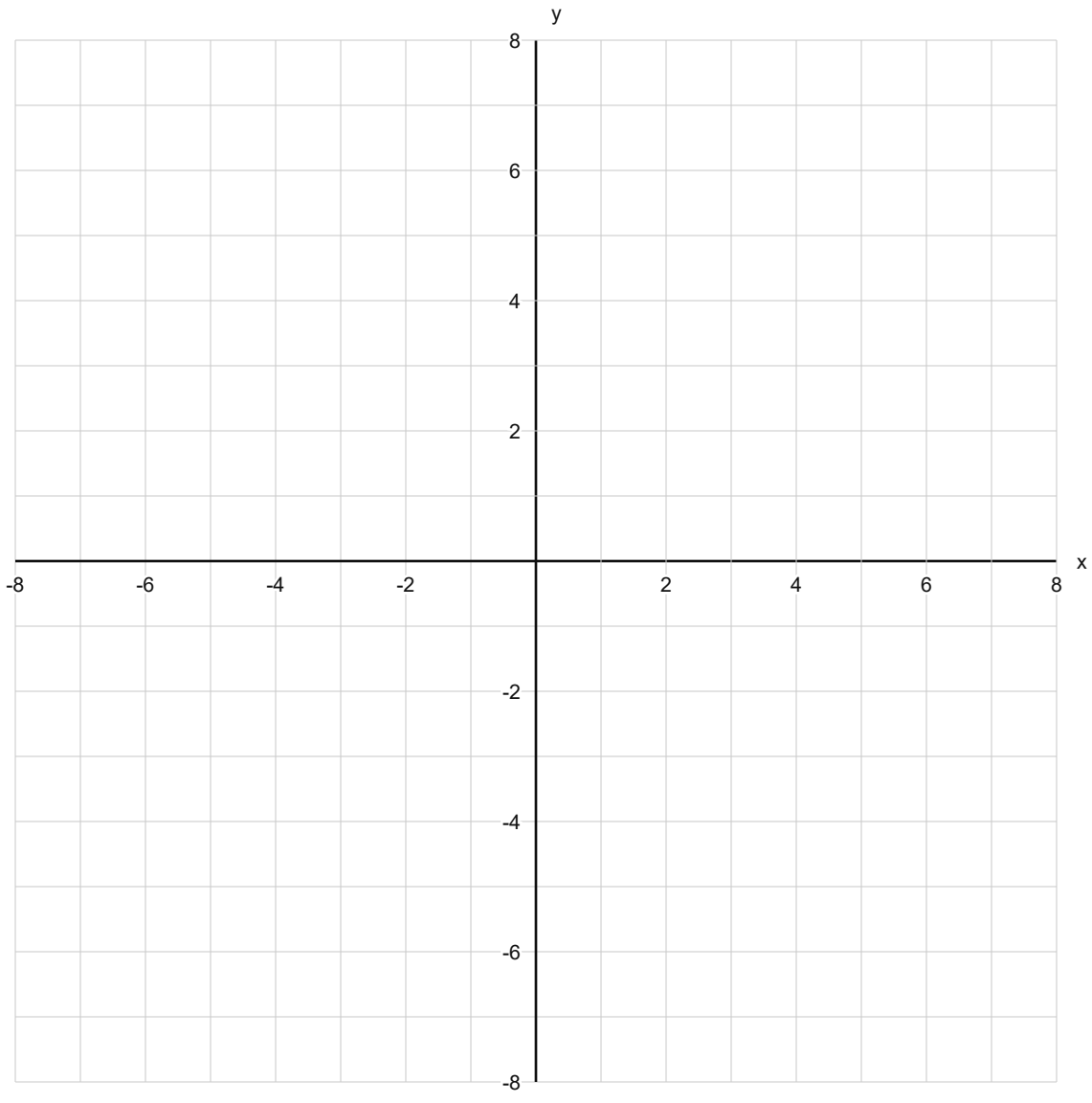
Algebra
x-intercept(s)
y-intercept
Hole(s)
Vertical asymptote(s)
Horizontal or slant asymptote(s)

Continue with part (b) on the next page.

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5. (continued)

(b) Sketch the function on the grid below and label the features you found.



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6. 4 points

Let $S(x) = \dfrac{-x^2 + 5}{2x - 4}$.

(a) Find all horizontal or slant asymptotes and justify them algebraically. For each type, give its equation or state that none exists.

Algebra	
Horizontal asymptote(s)	Slant asymptote(s)

(b) State the far-left and far-right end behavior. Explain algebraically why the graph approaches the asymptote.

Far-left behavior	Far-right behavior
Explanation	

7. 4 points

A workshop has made 10 kits at a total cost of \$180. Each additional kit costs \$12 to make. After n additional kits, the average cost per kit is $C(n) = \dfrac{12n + 180}{n + 10}$, for nonnegative integers n. Interpret the long-term average cost, including units. State whether it approaches that value from above or below and whether it reaches the value for any finite n. Justify your conclusions.

Answer and justification

Long-term value, units and meaning
Approach from above or below
Finite attainment and justification

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8. 4 points

Construct a rational function whose only x-intercept is $(-4, 0)$, whose only vertical asymptote is $x = -1$, whose horizontal asymptote is $y = 3/2$, and whose only hole is at $x = 2$.

(a) Write your function in factored form and verify that it has each required feature.

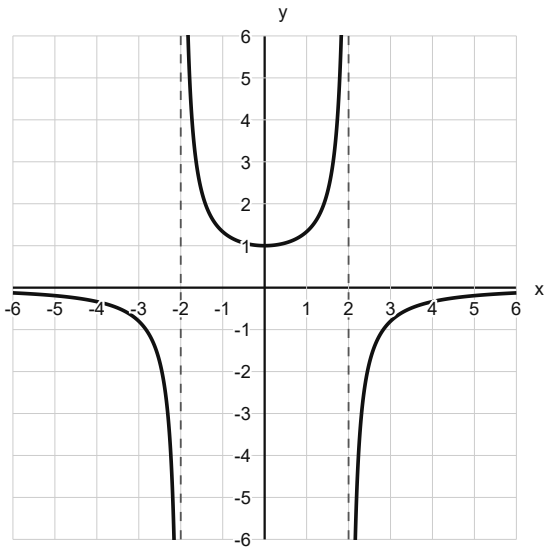
Function
Verification of the stated features

(b) Give the coordinates of the hole for your function. Show your calculation.

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9. 2 points

The graph represents a nonzero rational function. Which statement about its numerator and denominator degrees must be true? Select one answer and justify your choice.



A. The degrees are equal.

B. The numerator degree is exactly one greater than the denominator degree.

C. The numerator degree is exactly two greater than the denominator degree.

D. The denominator degree is greater than the numerator degree.

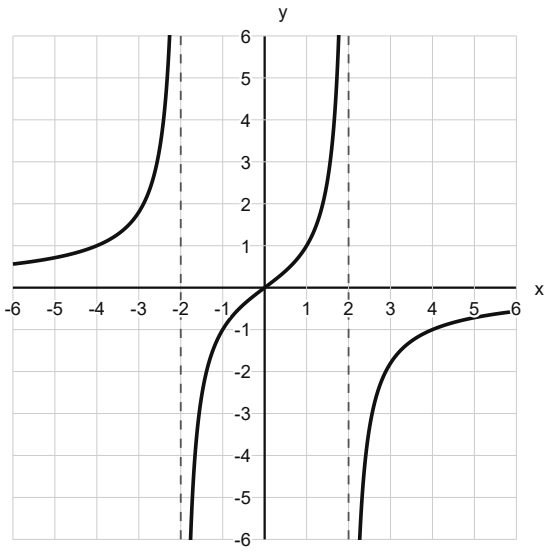
Answer

Choice	Justification

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10. 2 points

Which equation matches the graph? Select one answer. Justify your choice using the graph's behavior at the origin and at least one other relevant graph feature.



A. $y = \dfrac{-3x^2}{x^2 - 4}$

B. $y = \dfrac{-3x}{x^2 - 4}$

C. $y = \dfrac{-3x}{x^2 + 4}$

D. $y = \dfrac{3x}{x^2 - 4}$

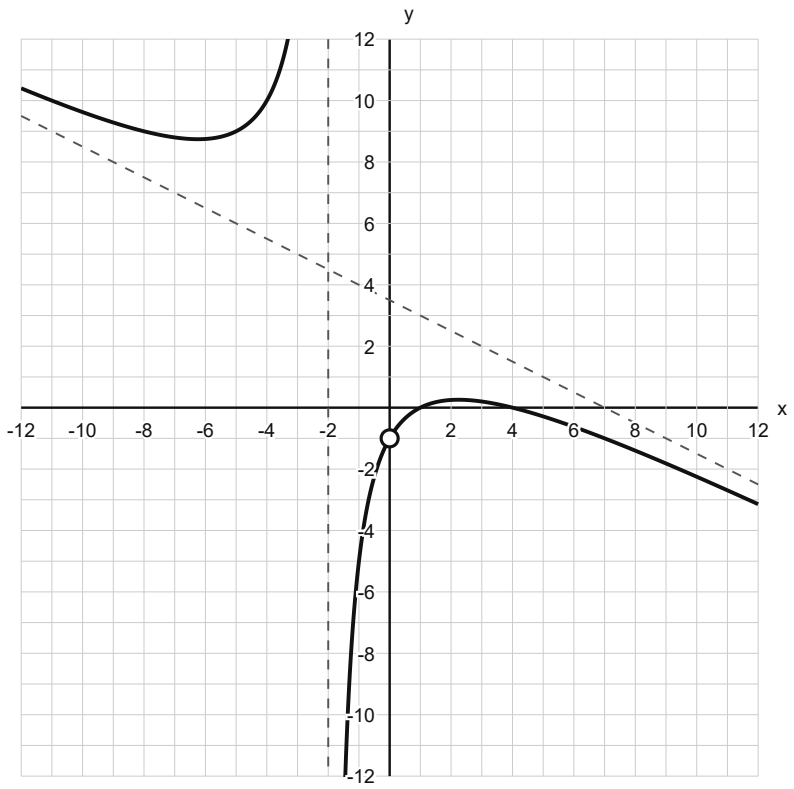
Answer

Choice	Justification

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11. 2 points

Which function matches the entire graph? The curve continues beyond the window. Select one answer. Include the missing point in your justification.



A. $y = \dfrac{-x(x-1)(x-4)}{2x(x+2)}$

B. $y = \dfrac{-(x-1)(x-4)}{2(x+2)}$

C. $y = \dfrac{-x(x-1)(x-4)}{2x(x-2)}$

D. $y = \dfrac{x(x-1)(x-4)}{2x(x+2)}$

Answer

Choice	Justification

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12. 2 points

Let $f(x) = -5x^6 + 2x - 1$ and $g(x) = \dfrac{3x^m - 8x^4 + 2}{x^2 + 1}$, where m is a positive integer. Which value cannot make the far-left and far-right end directions of g match those of f? Select one answer and justify your choice.

A. $m = 1$

B. $m = 4$

C. $m = 5$

D. $m = 3$

Answer

Choice	Justification

13. 4 points

A rational function H has only two undefined real inputs, 1 and 3. One is a hole and the other is a vertical asymptote. Selected values are shown; decimals are rounded. Based on the table, identify the likely x-coordinate of each and explain the evidence for each classification.

x	0.9	0.99	0.999	1	1.001	1.01	1.1
H(x)	-17	-197	-1997	undefined	2003	203	23

x	2.9	2.99	3	3.01	3.1
H(x)	4.0526	4.005	undefined	3.995	3.9524

(a) Hole

Location	Evidence

(b) Vertical asymptote

Location	Evidence